ResGate: Topic-Scoped Refusal Removal via Single-Layer Residual Gating

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Abstract

A transformer computes layer by layer through a residual stream, and that ordering can be put to work. We confine a rank-1 refusal-removing weight edit to a contiguous **band** of middle layers — 6 tensors at 9.4% of depth on Phi-4-mini — and read a binary gate from a layer **strictly below** that band. Every layer the gate reads is therefore left untouched, so the hidden state at the read site is **bit-identical whether or not the edit will fire**. The gate's decision is causally prior to the edit, and **one forward pass serves a conditional edit**: no second pass, no speculative generation, no router trained against the edited model. Band surgery is what creates the room. An edit spread over all depth would leave nowhere to read from.

The gate is closed form throughout — one direction and one threshold, computed from activations at one layer by subcategory-centred within-class scatter and smooth Tikhonov shrinkage, with no gradients, no search and no learned router. On three architectures the edit removes refusal almost completely at low KL (0.0745–0.0905), and the gate withholds it: on every prompt where the gate does not fire the gated output is **byte-identical** to the unedited model, while the same edit applied unconditionally changes every one of those outputs. The gate routes **78%, 81% and 82%** of 675 fresh out-of-corpus cyber prompts to the edit (lifting judged compliance on a 64-prompt fresh cyber sample to **85.9% on all three models**) and fires on **7%, 9% and 7%** of 234 non-target harmful prompts. On the prompts the unedited model refused, the gate bit and the judged outcome agree on **294 of 294**: nothing was answered without the gate firing, and nothing that fired was refused. The same edit with no gate answers 100% of those prompts, so the gate withholds a capability that is fully present. A second target category, bio, reaches **98.9–99.4%** coverage (**127/127 = 100%** on held-out validation rows) at **0.0–0.4%** leak onto the cyber set and **0/637** onto same-corpus cyber prompts.

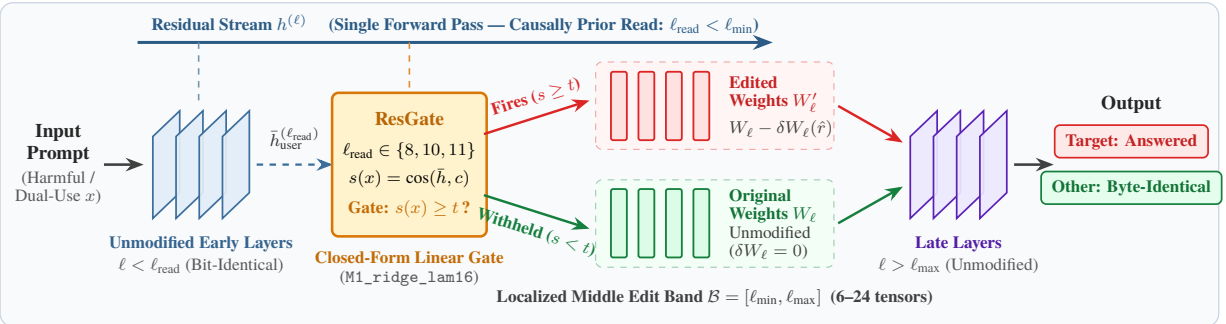


Figure 1: **ResGate single-pass architecture.** An input prompt passes through unmodified early layers to **ResGate**, which reads a single upstream layer prior to the middle edit band and decides in closed form whether the prompt belongs to the target topic. When the gate **fires**, rank-1 refusal-removing weight edits are applied across the middle band; when **withheld**, computation proceeds through the **original weights**, producing an output byte-for-byte identical to the unedited model in a single forward pass.

1 Introduction

Abliteration removes a language model's refusal behaviour with a rank-1 orthogonalising weight edit [4]. Doing so for one harm category while preserving refusal elsewhere has been approached by shaping the contrast dataset, which leaves the edit permanent and always-on. We make the edit conditional instead, and the mechanism is a **separation of layers**.

A decoder-only transformer writes into a residual stream layer by layer. Confine the edit to a contiguous band of middle layers and every layer *below* the band is left untouched — not approximately, but bit for bit, in both arms. A linear classifier read from one of those layers therefore sees the same hidden state whether or not the edit will fire downstream. Its decision is causally prior to the edit, so it can be taken once at prefill and the edit applied, or withheld, in the *same* forward pass. Three things are doing work and they are not interchangeable: the architecture is what *permits* this, band surgery is what *creates the room*, and the gate is what *uses* it.

That ordering also makes the design verifiable in a way a conditional intervention usually is not. If the gate's input cannot depend on the edit, then on a prompt the gate declines the served output must be the unedited model's output exactly — and it is, byte for byte, on every withheld prompt of two harmful test sets and three architectures (§6.3). The literature's verdict on gates is that the gate is the fragile part [11, 5, 33]; byte identity is the strongest answer we can give to it.

1.1 Contributions

- **Route-invariant read-early, edit-late gating.** Reading the gate strictly below the edit band ensures that h at the read site is identical whether or not the edit fires, so a single forward pass serves a conditional weight edit with exact byte identity on withheld prompts (§2.4; building on the early-read routing pattern of Residual Paving [11]).
- **A closed-form ridge gate.** Residual Paving [11] fits its router by gradient descent with weighted BCE and identifies learned route selectivity as its primary bottleneck (an oracle router gains a median $+12.9$ pp across six backbones, up to $+50.7$ pp). Our gate direction, edit band, and threshold are instead computed in closed form from activations via subcategory-centred scatter and Tikhonov shrinkage (§2.3): no gradients, no search, no learned router, and no random seed.
- **Gated rank-1 orthogonalisation.** Coupling a pre-band linear gate with a rank-1 row-normalised weight edit yields targeted compliance on the selected category while leaving non-target refusals intact, combining three properties that have previously appeared only in pairs (§3).
- **Live end-to-end verification.** Across three architectures and two target domains (cyber and bio), we verify exact byte identity on withheld prompts and a 294-of-294 agreement between the gate bit and the judged outcome on refused prompts.

Status of the figures in this paper. Every figure below is measured and traceable to a record directory (§9). Throughout, **LIVE** means served by our own HTTP server with one real forward pass per prompt and **OFFLINE** means scored from stored activations; where a quantity was not measured we say so and print no number for it.

2 Method

2.1 The rank-1 residual-stream edit

For each matrix W that **writes to the residual stream** — `self_attn.o_proj`, `mlp.down_proj`, and `experts.down_proj` on MoE models — and a unit refusal direction v in output space (orthogonalised against the harmless centroid following [4, 14]), the rank-1 PRE row-normalised update is $W' = W - w\,(n \odot v)\,u^\top$ with $n = \text{rownorm}(W)$ and $u = v^\top(W/n)$. Row-normalising before forming the pullback keeps the correction uniform across output rows of very different scale (reproducing the unconstrained reference implementation [14] to a cosine of 0.99985).

Rather than materialising modified weights on disk, we serve the equivalent forward hook

$$y' = y - w\,(x \cdot u)\,(n \odot v), \tag{1}$$

which matters for §6.3: on a prompt the gate withholds, this subtracts nothing, so the computation is the unedited model's own and the base weights are never touched. Query, key and value projections are left alone — they read the residual stream, they do not write it.

The single-direction premise the edit rests on is Arditi et al. [4]; for published arguments that refusal is not exhausted by one direction see Joad et al. [15], Wollschläger et al. [31] and Piras et al. [23].

2.2 Localising the edit band

To place the gate strictly below the intervention, we first localise the edit to a contiguous mid-network band. The direction at layer l is the difference of means of harmful and harmless activations at the last prompt token, orthogonalised against the harmless centroid; on why a pure through-origin direction can be the wrong object see Marshall et al. [19]. The **dose weight** is an effect size, not a raw gap:

$$\text{sep}(l) \;=\; \frac{\left\|\text{mean}_i(h_{i,l}) - \text{mean}_j(g_{j,l})\right\|}{\text{mean}_j\left\|g_{j,l} - \bar{g}_l\right\|}, \tag{2}$$

the harmful-to-harmless gap measured **in units of the harmless cloud's own radius**. Dividing by the spread removes the part of the gap that merely reflects a layer having large activations. The weights are normalised to mean 1, so the scalar K sets total edit energy and $\text{sep}(l)$ only redistributes it: $w_l = K \cdot \text{sep}(l) / \text{mean}_l\,\text{sep}(l)$. The quantity to dose-match across models is not K but $\max|\Delta W|$, which is linear in K.

The band is **measured per model**, never assumed: compute $w(l)$ for every layer from one capture, pooling at the last prompt token; admit the layers with $w(l) \geq 0.9 \cdot \max_l w(l)$; take the contiguous hull of that set. The bands so found span 7 of 30, 3 of 32 and 12 of 32 layers — **23.3%, 9.4% and 37.5% of depth**. The ratio of largest to smallest w across depth is ≈ 3.6, 2.43 and 2.17.

The rule finds a peak on two models and a plateau on the third. Phi's w is 1.481, 1.482, 1.393 against a model maximum of 1.482 — unambiguous, with the peak at L14. Llama's sits between 1.10 and 1.29 from L11 all the way to L27, so the admitted set is non-contiguous ($\{11, 12, 13, 15, 16, 20, 21, 22\}$) and the hull adds four layers purely to close it. We report Llama's band as a weakness of the model, not of the rule, and "surgical" is a far weaker claim at 37.5% of depth than at Phi's 9.4%. Contiguity is a specification convenience, not a requirement: the implementation subsets the band at run time, so any set of layers works.

What the band buys is not only KL. A narrow band is usually motivated as a cheaper edit. Its load-bearing property here is different: it leaves a majority of the depth — and in particular the layers below it — *unmodified*, which is the only reason a read can be placed where the edit cannot reach it. Where refusal lives in depth is contested, and we cite that rather than settle it: Zong et al. [37] find the block spanning layers 8–11 selected first in all six of their greedy searches; Eswar et al. [7] find a *"redundant subspace"* admitting many disjoint sufficient masks; Li et al. [18] read a contiguous mid-band the other way, as *necessary* "safety

layers"; and Hase et al. [9] show that causal-localisation conclusions *"do not provide any insight into which model MLP layer would be best to edit"*.

2.3 The gate: how it is extracted

The gate is **binary**: one direction, one threshold, one bit, frozen at prefill. For a target category against all other harm categories the between-class scatter has rank $C - 1 = 1$, so one direction is the correct object rather than a simplification.

Notation. All rows are pooled over the user-content span only (`mean_user`) at read layer ℓ, and **each row is unit-normalised before any mean**: X_p are the TRAIN rows of the target category; X_n the REF rows, 10 drawn from each non-target harm category; $p = \text{mean}(X_p)$ and $g = \text{mean}(X_n)$; $d = p - g$; and $n = \text{rows}(X_p) + \text{rows}(X_n)$.

Subcategory-centred within-class scatter. The raw difference of means $d = p - g$ points partly along directions that are wide *within* each class — nuisance variance such as prompt length, syntactic framing and register. Because the reference set $X_n = \bigsqcup_{s=1}^{S} X_{n,s}$ spans S distinct non-target harm categories ($S = 14$ categories of $n_s = 10$ rows each on the 240-row fit), centring X_n on a single pooled mean $g = \text{mean}(X_n)$ would leave the $S-1$ between-subcategory offsets $(\bar{g}_s - g)$ inside the scatter matrix, where $\bar{g}_s = \text{mean}(X_{n,s})$:

$$\Sigma_{\text{global}} = \Sigma_{\text{subcat}} + \frac{1}{n-2} \sum_{s=1}^{S} n_s (\bar{g}_s - g)(\bar{g}_s - g)^\top. \tag{3}$$

Those $S - 1$ inter-category offsets inflate the leading covariance modes and crowd out true within-category nuisance variance. We therefore centre the target class on p and each negative subcategory $X_{n,s}$ on its own centroid $\bar{g}_s$, removing between-subcategory scatter by construction while keeping the mean-difference signal $d = p - g$ unchanged:

$$\begin{aligned} Z_{\text{subcat}} &= \big[\, X_p - p \;;\; X_{n,1} - \bar{g}_1 \;;\; \ldots \;;\; X_{n,S} - \bar{g}_S \,\big], \\ \Sigma_{\text{subcat}} &= \frac{Z_{\text{subcat}}^\top Z_{\text{subcat}}}{n - 1 - S}. \end{aligned} \tag{4}$$

Note that Σ_{subcat} is **within-class**, not total, covariance — this is the LDA object, not PCA. Centring each class and subcategory on its own mean first removes the between-class signal, so the whitening suppresses nuisance rather than the thing we want to read.

Smooth Tikhonov whitening via the Gram spectrum. Rather than truncating the covariance spectrum at a hard top-k step (which leaves 29–50% of $\|d\|^2$ across modes $6 \ldots 15$ unwhitened), we apply smooth Tikhonov shrinkage anchored at an eigenvalue scale $\alpha = \lambda_{k+1}$ of Σ_{subcat}:

$$\begin{aligned} c &= \text{unit}\big(\alpha(\Sigma_{\text{subcat}} + \alpha I)^{-1} d\big) \\ &= \text{unit}\Big((I - VV^\top)\, d + V\, \text{diag}\Big(\frac{\alpha}{\lambda_i + \alpha}\Big) V^\top d \Big), \end{aligned} \tag{5}$$

where (λ_i, v_i) are all positive eigenpairs of Σ_{subcat}. Because $\text{rank}(\Sigma_{\text{subcat}}) \le n - 1 - S$, the $d \times d$ matrix is never formed; the eigenvectors V are lifted in closed form from the $n \times n$ Gram matrix $Z_{\text{subcat}} Z_{\text{subcat}}^\top$, whose symmetry and positive semi-definiteness make the lifted columns orthonormal — load-bearing, since $VV^\top$ is a projection only then. Every positive variance mode is attenuated smoothly by $\alpha/(\lambda_i + \alpha) \in (0, 1)$, hardest on the widest nuisance directions, while the orthogonal nullspace ($\lambda_i = 0$) stays at weight 1: components

are **shrunk, not deleted**, so signal lying along a noisy axis is retained at reduced weight. We set $\alpha = \lambda_{16}$ (`M1_ridge_lam16`) throughout, spanning the $S - 1 = 13$ subcategory degrees of freedom plus the leading style axes with zero learned parameters, gradients or search.

The threshold. L is a **declared leak budget**, not a measured property. With $m = \lfloor L \cdot \text{rows}(X_n) \rfloor$, two rules:

$$\text{R2:} \quad t = \min\{\, s \in \textsc{train} : \#\{X_n \geq s\} \leq m \,\} \tag{6}$$

$$\text{R3:} \quad t = r_{m+1} + \varepsilon \tag{7}$$

where r are the sorted REF scores. R2 can only place t on one of the TRAIN scores, and we show a case where the operating point we wanted fell in a **gap in that staircase** — 0.0323 of score with no candidate in it. R3 has a finer grid and cannot become vacuous, because the constraint itself sets t. ε **must exceed the model's measured live-versus-offline reproducibility bound**: a one-ULP version failed, its binding certificate surviving neither live scoring nor a change of BLAS reduction order (§8).

A stated rule can also fail to be the thing that produced the threshold. If $\min(\textsc{train}) > r_{m+1}$ then every candidate was feasible, t fell onto the lowest TRAIN score, and $\#\{X_n \geq t\} = m$ is a coincidence: we call such a cell VACUOUS, report the distinction for every cell, and do not ship one.

At serve time. $\text{score}(x) = \langle x/\|x\|, c \rangle$, fire iff score $\geq t$. The decision is taken once at prefill, stored as a per-sequence boolean, and not revisited per token.

2.4 Choosing the read layer

Two constraints, one hard and one empirical.

Hard: the read must be strictly below the edit band. $\max(\text{read}) < \min(\text{edit})$ is asserted at construction and an in-band read is refused. This is the constraint the title calls *before* the edit: a lower layer index means the read happens earlier in the forward pass. Everything in §6 rests on it — every layer the gate reads is unmodified, so h at the read site is bit-identical in both arms, the decision cannot depend on the edit, and the edit can be applied to the same pass that produced the decision. The selected read layers are **L8, L11 and L10**, 2, 2 and 1 layers below each model's band, and each model uses that same single read layer for both the cyber and bio gates. That harm detection sits upstream of refusal execution is published as anatomy: Chu et al. [6] characterise a safety circuit in which *"Harmful Detection Heads"* precede the *"Refusal Heads"* that turn the signal into a refusal.

Empirical: select on out-of-corpus data, by the operating statistic. Within the admissible layers we select on a held-out half of an independently assembled corpus, ranking layers by **leak at matched coverage**. Two rules we explicitly reject. *AUC*: it is not the statistic the gate operates on, and it rises nearly monotonically with depth, so an AUC argmax always returns the deepest admissible layer. *Any in-corpus argmax*: selecting Phi's read layer by leak at matched coverage over 100 paired in-corpus resplits chose L5 over L11–L12 by $+3.45$ leaks per 378, bootstrap $[+2.72, +4.16]$, Wilcoxon $p = 4.6 \times 10^{-12}$ — and on a fresh corpus the sign reversed at every point tested, the out-of-corpus interval at matched coverage being $[-21, -1]$, which does not overlap. More seeds could not have helped: every resplit carries the same ratio of dimensions to rows, so resplit stability is structurally blind to this class of bias. The shipped layer on Phi is L11 (with L12 also verified live on the bio gate). The full account is in the long version of this paper and in `records/phi-oocsweep-2026-10-06/`.

Where the depth trend sits in the probing literature. Linear separability rising with depth is the canonical probing finding [2], though the richest representations are reported at mid-depth [27]; prompt domain is decodable from prefill hidden states and has been used to route between models [12]; and for adversarial intent specifically, *"surface form appears early, while pragmatic intent consolidates later"* [21].

Table 1: The six properties against the nearest published and community work. Y = yes, N = no, P = partial. Qualifications that would not fit a cell are given in the notes below, verbatim from the draft.

	1	2	3	4	5	6
Hadetskyi et al. [8]	Y	**N**[a]	Y	P[b]	Y	P
CAST [17]	P	P[c]	**N**	N	P	Y
Residual Paving [11]	N[d]	**Y**[e]	N[f]	P	**N**[g]	N
`apostate/diode` [3]	Y	**N**[h]	N[i]	P	Y	N
RepIt [26]	P	**N**[a]	N[j]	N	Y	Y
Palette [29]	N[k]	**N**[l]	P	N	Y	Y

[a] no gate at all. [b] spread over 25–95% of depth. [c] condition at L7; no edit, so no "below". [d] learned. [e] router at layers 1–4, ahead of a later-layer edit. [f] added bottleneck experts. [g] a *safe reframe*, which its authors state is not a direct-compliance target. [h] gate co-located inside the edited layers. [i] gated MLP neuron. [j] activation steering, not a weight edit. [k] multi-objective search. [l] no runtime gate.

3 Related work

Prior building blocks. Our design connects several lines of prior work:

- **Domain-specific refusal removal.** Hadetskyi et al. [8] remove refusal on Kimi K2 for the cybersecurity domain via multi-objective search without a runtime gate; Palette [29] selectively relaxes refusal on authorized target domains while preserving safety elsewhere.
- **Terminology.** The phrase "conditional directional abliteration" is used in community code (`apostate/diode`, [3]); we say **gated** to denote a single pre-band routing decision over a rank-1 weight edit.
- **Runtime gating.** CAST [17] reads a condition at layer 7 for activation steering; Residual Paving [11] routes from layers 1–4 and intervenes at later layers; CLEAR [30] drives a safety adapter from a hidden-state gate; SafeMoE [10] routes with a learned gating network; and Tripwire [36] realises its clamp as a detector-gated intervention.
- **Contrast-set composition.** Petrov et al. [22] show that the composition of the contrast baseline determines whether a refusal direction separates cleanly.

Six properties, counted generously (a partial counts as a half): (1) closed-form direction; (2) gate **below** the edit; (3) rank-1 weight edit; (4) narrow band; (5) compliance; (6) category routing. Nothing published reaches five (Table 1).

The gap is properties 2 + 3 + 5 together. Each pair exists. Hadetskyi et al. [8] score highest and have no gate at all. Residual Paving [11] is the closest on architecture and deserves the credit for it: it forms a boundary feature from residual states in $L_R = \{1, \ldots, 4\}$ and intervenes at later layers L_I, which is the read-early/edit-late separation this paper is built on. It differs in what is gated and how the gate is obtained. What is gated is an added set of prompt-conditioned bottleneck residual experts aimed at a *"safe reframe"* that its authors state is **"not a direct-compliance target"**, with the backbone left unchanged — rank-1 orthogonalisation is

absent from the paper. How the gate is obtained is gradient descent: the router is fitted *"to the edit-versus-keep boundary on cached early residual states with weighted BCE"*. Their own diagnostic then makes that router the headline: substituting a held-out oracle label for the learned scalar gate raises their keep-side score on *every* one of six backbones, by a median $+12.9$ pp and as much as $+50.7$, and learned benign preservation ranges from 57.2% to 97.9% across backbones while oracle routing reaches $\geq 99.9\%$ on all of them. They conclude that *"learned route selectivity remains the limiting factor"*. That conclusion is the motivation for the present paper: we keep their layer separation and replace the trained router with a closed-form one. We report no head-to-head — the backbones, the edit sets and the metrics all differ — so this is a difference in construction, not a demonstrated improvement. CAST [17] has gate and category but **"no weight update"**, so nothing exists for the gate to sit below. Two further near neighbours: LoMC [13] matches our operator — a rank-one layer-wise correction applied only inside a compact edit support — but its *"support-gated"* gate is an offline mask over *weights*, *"updates outside $\mathcal{S}$ are set to zero"*, not a read of the prompt, and its direction is a single harmful-versus-benign one rather than a category; and Alagharu et al. [1] do pair per-category refusal directions with *"a lightweight probe that determines whether to steer"*, but the categories exist only because the model was fine-tuned to emit categorical refusal tokens, the intervention is a steering vector, and the direction of travel is *increasing* refusal.

There is also a published argument against the design, and it is explicit. C-$\Delta\Theta$ [16] confines a weight edit in order to *eliminate* the gate rather than to enable one: *"conditional variants improve selectivity by gating when steering is applied but still retain an inference-time control path"*, and what it ships instead is *"no inference-time hooks, no per-request gating"*. Three further results say the gate is the fragile part: Residual Paving [11] (*"the oracle route is non-deployable"*), Bosco and Srinivasan [5] (a detector-gated intervention *"only matches a trivial rule"*), and Xun [33] (a calibrated gate OOD-inverts and *"silently reduces the gated pipeline to its base model"*). We take that as the standard this design has to meet, which is why §6 is a verification section rather than a table of rates.

Table 2: The three models, the edit each receives, and the result of applying that edit *unconditionally* — same 100 prompts, same judge, same decoding, no gate. The read layer is strictly below the band in every row, which is what §2.4 requires. OFFLINE.

model	arch	params	layers	hidden	edit band	read	tensors	K	original refused	edited refused	KL
`gemma-4-26B-A4B-it`	MoE	26B	30	2816	L10–16	L8	21	2.0	97/100	**1/100**	**0.0849**
`Phi-4-mini-instruct`	dense	∼4B	32	3072	**L13–15**	L11	**6**	2.0	99/100	**0/100**	**0.0745**
`Llama-3.1-8B-Instruct`	dense	8B	32	4096	L11–22	L10	24	1.5	94/100	**1/100**	0.0905

Full identifiers: `google/gemma-4-26B-A4B-it`, `microsoft/Phi-4-mini-instruct`, `meta-llama/Llama-3.1-8B-Instruct`. The KL column mixes two harnesses and always did; the ordering it appears to give is not one we claim (§5).

4 Experimental setup

Models. Three architectures, in Table 2. Gemma is the only MoE, so 7 of its 21 tensors are fused `experts.down_proj` stacks; the implementation proves its reconstruction of each stack's input against that stack's own forward before hooking it and declines any stack exceeding tolerance, because a mis-indexed fused edit is a wrong edit rather than a smaller one — Shi et al. [25] report that module-name matching drops such stacks and *"nothing raises an error"*. No weight is modified on disk in any experiment; the edit is applied as forward hooks on unmodified weights, verified byte-identical before and after by shard hash, inode and link count.

Corpora. No prompt text is reproduced anywhere in this paper (§9). The FIT set is `pool681`: 1426 rows assembled from advbench / eval177 / `harmful_behaviors` test — 692 harmful over 15 categories, 554

harmless, 180 excluded paraphrase rows. The OOC set is 3216 fresh rows: CySecBench 1350 (cyber), CatQA 469 (10 general harm categories), plus ICS, imperative and replication arms and 40 convention-control rows. A third set, HarmBench [20] with per-category labels, exists for Gemma only (197 rows). Disjointness between FIT and OOC: **0 exact matches** at three normalisations, token-set Jaccard nearest neighbour **max 0.438** with nothing at ≥ 0.5. The bound is stated: exact matching cannot catch a low-overlap paraphrase. Every fresh capture carries a convention control — a subset of FIT rows re-captured and compared against the stored activations, at cross-run distances of 4.12e-05, 1.38e-05 and 1.77e-05 against within-run bf16 floors of 7.44e-05, 1.21e-05 and 1.93e-05, with a wrong-layer sentinel $1350\times$ and $4517\times$ those distances on the two dense models. Without this, a convention drift would present as a model difference.

Judging. A three-way judge — REFUSED / DEFLECTED / COMPLIED — on an independently served model, rubric fixed across every arm. Three-way-or-finer response taxonomies are established [24, 35]. A keyword refusal count cannot distinguish *"I won't help"* from *"I won't help, but here is how you would do it"*, and those are different outcomes. That failure has been measured: keyword refusal precision of 11.2% across six abliterated models [34]; a Spearman correlation of -0.394 between string matching and human rankings [28]; substring ASR moved by up to 30 points by the generation budget alone [20]. Such marker lists are nonetheless the field's default [32]. The judge is probed before use on a clear refusal, a clear compliance and a clear deflection; judge noise, measured on byte-identical text across two runs, is **1 label in 138**.

A reply either refuses or it answers. The judge reports three labels, but the headline counts two states: *answered* means the reply is not a refusal. The prior work's keyword detector cannot make the third distinction and scores an evasive reply as a successful removal, so the two-state reading is the comparable one. We apply it identically to the cyber and non-target harmful sides — it helps us on one and hurts us on the other — and we publish the three-way counts so anyone can recompute.

We count only the prompts the original model refused. The original model already answers 16/64 Gemma, 16/64 Phi and 13/64 Llama CySecBench prompts on its own, because CySecBench phrases requests analytically. On those prompts the edit has nothing to remove, so they cannot show its effect. Earlier drafts subtracted a baseline; dropping the prompts is cleaner than correcting for them.

5 Results: the edit

With no gate, the localized middle-band edit removes refusal almost completely across all three architectures at low KL divergence (0.0745–0.0905; Table 2 and Table 3):

- **Minimal depth footprint:** Phi removes refusal completely (0/100 refused) by editing just **6 tensors (9.4% of depth)**, compared with 25–95% of depth (100% → 7%) in Hadetskyi et al. [8].
- **Three-way response split:** Because standard two-state scoring counts deflections as non-refusals, Table 3 reports the full REFUSED / DEFLECTED / COMPLIED breakdown (§6.3 shows our gate conclusion holds under either reading).
- **KL harness sensitivity:** Re-measuring Gemma at batch size 8 (matching Phi/Llama's driver) shifts first-token KL from 0.0849 to 0.0945 ($+11.3\%$, flipping the Gemma–Llama order while leaving keyword refusal unchanged at 33/100); accordingly, we do not rank Gemma against Llama by KL.

Table 3: Unconditional middle-band edit breakdown and harness controls ($n = 100$ prompts; `records/abl-control-2026-10-07/`).

Metric	Gemma (L10–16)	Phi (L13–15)	Llama (L11–22)
Edited tensors (% depth)	21 (23.3%)	**6 (9.4%)**	24 (37.5%)
REFUSED / DEFLECTED / COMPLIED	1 / 30 / 69	**0 / 5 / 95**	**1 / 3 / 96**
First-token KL (batch 20 / batch 8)	0.0849 / 0.0945	— / **0.0745**	— / 0.0905
Hooked tensors unreached	0 / 21	0 / 6	0 / 24
Always-on $\neq$ original outputs	100 / 100	100 / 100	99 / 100

6 Results: the gate confines the edit

Three versions of each model answer the same prompts: **original**, no edit at all; **always-on**, the edit applied to every prompt with no gate; and **gated**, the classifier deciding prompt by prompt.

Table 4: The complete picture. Two gates (`M1_ridge_lam16`, $\alpha = \lambda_{16}$), three prompt sets, three architectures; cells read Gemma (L8) / Phi (L11) / Llama (L10). The diagonal (bold) is coverage, the rest is leak. Counts first, rate second. Every cell LIVE.

Gate	Threshold t (`op["thresh"]`)	Cyber (675 CySecBench)	Bio (637 WMDP_BIO)	Non-target Harm (CatQA)
Cyber gate	.1135 / .0530 / .0689	526 / 549 / 551 = **77.9 / 81.3 / 81.6%**	5 / 2 / 33 = 0.78 / 0.31 / 5.18%	16 / 22 / 17 = 6.8 / 9.4 / 7.3%
Bio gate	.0580 / .0422 / .0852	3 / 0 / 2 = 0.44 / 0.00 / 0.30%	633 / 630 / 632 = **99.4 / 98.9 / 99.2%**	15 / 8 / 10 = 6.4 / 3.4 / 4.3%

Non-target harmful is the CatQA TEST-half aggregate excluding each gate's own target category ($n = 234$; Phi 235). Exact float64 thresholds t: cyber 0.1134911512 (Gemma L8, `iso_leakage`), 0.0530322825 (Phi L11, `iso_leakage`), 0.0689013176 (Llama L10, `in_fit_R3`); bio 0.0580110430 (Gemma L8), 0.0422080616 (Phi L11; 0.0419398590 at L12), 0.0852456303 (Llama L10), all `R3U140`. Clopper–Pearson 95% CIs — cyber coverage: $[74.6, 81.0]$, $[78.2, 84.2]$, $[78.5, 84.5]$; bio coverage: $[98.4, 99.8]$, $[97.8, 99.6]$, $[98.2, 99.7]$. Records: `records/cyber-m1-live-serve-2026-10-15` and `records/bio-m1-live-serve-2026-10-15`.

6.1 Firing and live generation compliance

Table 4 (cyber row), Figure 2 and Table 5 summarise the cyber gate across all three architectures:

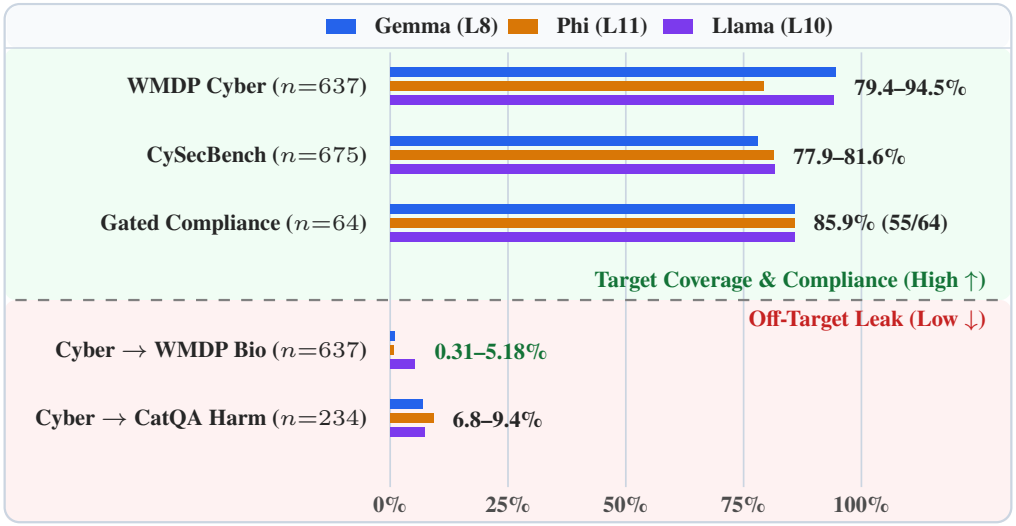


Figure 2: Cyber gate coverage, compliance and off-target leak (LIVE). Gated compliance reaches 55/64 (85.9%) across all three models (up from 20.3–25.0% unedited).

Table 5: Cyber gate live generation compliance (64-prompt fresh cyber subset) and live-vs.-offline numerical fidelity across 2,875–2,876 rows (`M1_ridge_lam16`). LIVE.

Metric	Gemma (L8)	Phi (L11)	Llama (L10)
Base compliance (unedited)	16/64 (25.0%)	16/64 (25.0%)	13/64 (20.3%)
Gated compliance (live)	**55/64 (85.9%)**	**55/64 (85.9%)**	**55/64 (85.9%)**
Share of forced-edit ceiling	84.8%	84.8%	85.7%
Benign control compliance	10/10 (100%)	10/10 (100%)	10/10 (100%)
WMDP_CYBER test ($n = 637$)	602 (94.5%)	506 (79.4%)	600 (94.2%)
Live vs. offline agreement	99.86% (4)	99.93% (2)	99.90% (3)
Mean / max $\lvert\Delta_{\text{score}}\rvert$	4.2e-4 / 6.9e-3	1.5e-4 / 1.0e-2	1.7e-4 / 1.0e-3
P10 24-row replay max $\lvert\Delta\rvert$	0.0 (24/24)	0.0 (24/24)	0.0 (24/24)

6.2 On refused prompts, the gate bit is the whole story

Restricting to prompts the original model refused on the 138-prompt served subset isolates whether the edit changed the reply (Table 6): **294 prompts, 294 agreements.** The two middle columns are empty (115 fired and answered; 179 held and refused).

Table 6: Gate bit against judged outcome, on prompts the original model refused (`M1_ridge_lam16`). The two middle columns are empty. LIVE.

		fired		held	
	n	answered	refused	answered	refused
Gemma cyber	22	15	0	0	7
non-target harmful	58	5	0	0	53
Phi cyber	45	39	0	0	6
non-target harmful	58	7	0	0	51
Llama cyber	50	43	0	0	7
non-target harmful	61	6	0	0	55

6.3 The strongest evidence is byte identity

On every prompt where the gate does **not** fire, the served output matches the unedited model byte-for-byte via `sha256` (Table 7), and across all 113 live 3-arm `/compare` and `/generate` checks it matches the forced-edit branch byte-for-byte whenever the gate fires.

Table 7: Byte identity on withheld harmful prompts (LIVE).

Model	gated = original (sha256)	always-on ≠ original
Gemma	9/9 (cyber) and 58/58 (CatQA)	67/67 changed
Phi	10/10 (cyber) and 57/57 (CatQA)	67/67 changed
Llama	8/8 (cyber) and 58/58 (CatQA)	66/66 changed

6.4 Two bookkeeping notes

Continuous R3 vs. discrete R2. R2 restricts t to the 100 TRAIN scores, which can stall on a coarse tail gap (e.g. Phi $m = 14$ drops to 415/675 coverage at 15/235 leak), whereas matching the leak budget continuously

(`iso_leakage`, $t = 0.053032$) routes **549/675 (81.3%)** at 22/235 live leak; on Llama ($m = 17$), in-fit R3 and `iso_leakage` coincide to 3.4×10^{-5} (**551/675** and **17/234**).

Exact threshold sweeps. Because $\ell_{\text{read}} < \ell_{\text{min}}$, greedy gated output is a deterministic step function of score$(x) \geq t$, switching cleanly between the pre-scored unedited and forced-edit completions.

7 Results: a second target category

We apply the same closed-form recipe (`M1_ridge_lam16`, $m = 6$ `R3U140`) to biological threats and screen three additional candidate categories (Figure 3 and Table 8).

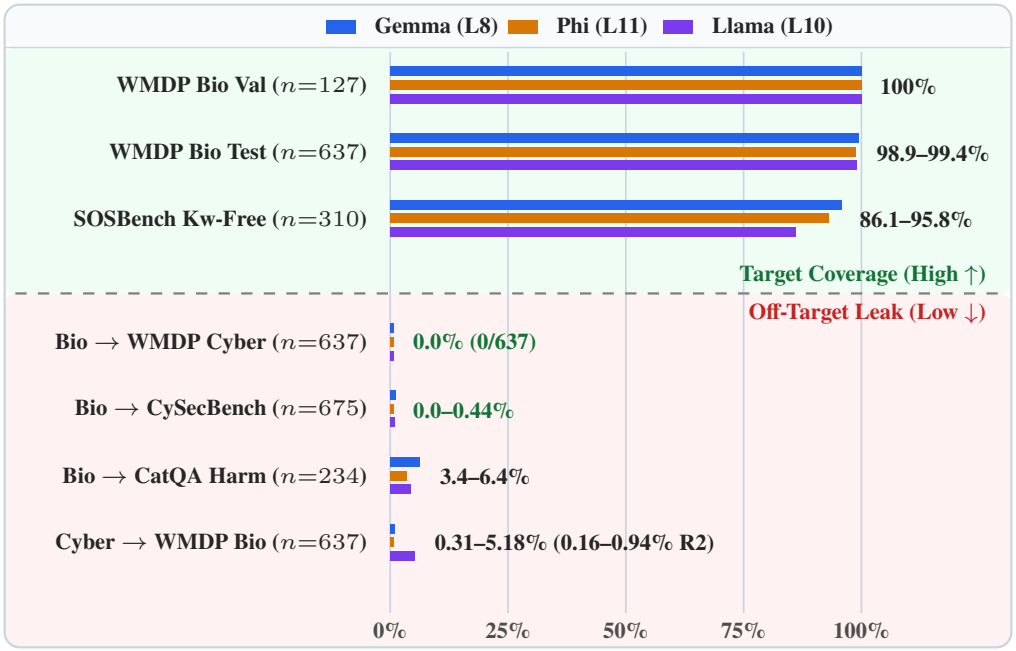


Figure 3: Bio gate selectivity across all three architectures (LIVE). Green zone: target coverage on held-out validation, test and keyword-free external benchmarks. Red zone: cross-domain leakage onto cyber and non-target harmful benchmarks.

Table 8: Second target category (Bio gate, `M1_ridge_lam16`, $m = 6$ `R3U140`) and candidate category screening. Cells report Gemma (L8) / Phi (L11) / Llama (L10).

Evaluation Surface	Gemma (L8)	Phi (L11)†	Llama (L10)
A. Bio Gate Target Coverage (LIVE)			
WMDP_BIO val ($n = 127$, 0-overlap)	**100%** (127)	**100%** (127)	**100%** (127)
WMDP_BIO test ($n = 637$)	**99.4%** (633)	**98.9%** (630)	**99.2%** (632)
SOSBench Bio ($n = 498$, external)	**96.4%** (480)	**95.0%** (473)	**89.0%** (443)
SOSBench Kw-Free ($n = 310$, 0 stems)	**95.8%** (297)	**92.9%** (288)	**86.1%** (267)
B. Cross-Domain & Non-Target Leak (LIVE)			
Bio → WMDP_CYBER ($n = 637$)	**0.00%** (0)	**0.00%** (0)	**0.00%** (0)
Bio → CySecBench ($n = 675$)	0.44%(3)	**0.00%** (0)	0.30%(2)
Bio → PKU Cyber ($n = 612$)	3.59%(22)	1.14%(7)	0.49%(3)
Bio → CatQA ($n = 234/235$)	6.4%(15)	3.4%(8)	4.3%(10)
Cyber → WMDP_BIO ($n = 637$)‡	0.78%(5)	0.31%(2)	5.18%(33)
(at in-fit R2 threshold)	0.31%(2)	0.16%(1)	0.94%(6)
C. Live Serving Numerical Fidelity ($N = 3,265$–$3,266$ rows)			
Max $\lvert\text{live} - \text{offline}\rvert$	6.39e-03	5.48e-03	9.68e-04
Live vs. offline decision flips	3 / 3,265	**0 / 3,266**	2 / 3,265
D. Candidate Category Screening (Disqualified at $> 10\%$ Leak)			
Influence Ops ($n = 25$ / ∼210)	88%/ 22.5% leak	60%/ 19.5% leak	76%/ 15.3% leak
Fraud ($n \approx 25$ / ∼208)	73%/ 28.4% leak	79%/ 19.9% leak	81%/ 26.4% leak
Hate (ToxiGen vs. SALAD O2)	18% vs. 100%	18% vs. 62%	35% vs. 99%

†Phi bio gate at L12 ($t = 0.041940$): 100% val, 98.7% test, 95.8% SOSBench, 93.9% Kw-Free, 0% cyber leak, 4.7% CatQA leak. ‡Selection-clean (0 WMDP rows in cyber fit/threshold); 34 bio prompts fire any model, only 2 fire all three.

8 Limitations

Two corpora. Every interval above bounds row sampling within one corpus. Nothing here bounds corpus-to-corpus variation, and two corpora cannot. Only Gemma has a third.

A declared leak budget permits more than its label says. The budget is an integer count m over a finite reference set, so what it guarantees is an interval, not a rate. $m = 14$ on 140 reference rows has a Clopper–Pearson 95% upper bound of **16.21%**. The cyber budgets of $m = 11/14/17$ permit **13.6 / 16.2 / 18.7%** respectively. The cyber gate's observed 6.8–9.4% leak is therefore an outcome, not a guarantee of its budget, and a reader should not treat the declared L as a bound on the realised rate. The bio gate's $m = 6$ has an upper bound of 9.09% — the tightest budget in this paper, and the only one whose label survives the interval arithmetic.

Live decisions can differ from offline ones. On the 2875–2876 scored rows of the cyber gate sweep (`M1_ridge_lam16`), measured max $|\Delta|$ between served and stored scores is 6.92e-03 on Gemma, 1.01e-02 on Phi and 1.02e-03 on Llama (mean 4.21e-04, 1.51e-04 and 1.68e-04), producing 4, 2 and 3 decision flips across the entire 2875-row surface (99.86–99.93% agreement); on the 3265–3266 rows of the bio gate sweep, max $|\Delta|$ is 6.39e-03, 5.48e-03 and 9.68e-04, producing 3, 0 and 2 flips. All per-model jitter figures are **sample maxima rather than bounds** (§6.1). Relatedly, one-ULP threshold certificates are fragile at float precision: placing t a single ULP above r_{m+1} cannot immunise against 10^{-4} live `bfloat16` reduction drift unless a finite margin ε is added (§7). All served thresholds in `records/cyber-m1-live-serve-2026-10-15` and `records/bio-m1-live-serve-2026-10-15` are stored and verified at full IEEE-754 float64 precision.

Capability retention. This is measured on one model. Gemma has a blind, position-randomised pair-

wise arm over 50 prompts (McNemar $p = 1.00$ against the original). Phi and Llama have 3 and 8 benign prompts, and benign compliance *fell* on both (Llama $9 \to 6$, Phi $8 \to 7$, into DEFLECTED not REFUSED). KL cannot substitute: this project measured a build at half the KL judged *worse*.

Gemma self-judges. The judge is the same model family as the Gemma checkpoint under test. The Phi and Llama rows do not carry this.

A withdrawn claim. Our contrast with Hadetskyi et al. [8] rested on a result recorded as *"not reproduced. Zero selectivity."* An audit finds that arm differs from their method on six points, four of them load-bearing: a dense 8B rather than a 1T MoE, where scale is part of their stated mechanism; a 1-harmful-plus-4-harmless contrast kit rather than their prompt sets per domain; CBRN as the judged target rather than their cyber; and CBRN against fraud as the selectivity pair rather than cyber against explicit content, with no explicit-content prompt in the probe. Edit span and operator normalisation also differ. Independently, every arm in that sweep sat at a corner of the dose dial (0/0/0 or 88/92/82) while their claimed state is partial, so the design could not have detected selectivity had it been present. **All we assert is what was measured: a single-prompt domain-matched kit on Llama-3.1-8B is fully functional and completely non-selective —** $94/100/92\% \to 0/0/0\%$ **on 96 prompts. We make no claim either way about whether their result holds.** Record: `records/replication-audit-2026-10-07/`.

We do not claim that corpus-driven selectivity fails. We reproduced RepIt [26] on Phi-4-mini with their released code and zero source modifications: target ASR **0.8023** against their published **0.8136** (McNemar $p = 0.875$), held-out selectivity 3.55 baseline-adjusted. Their raw difference-of-means direction already discriminates at 1.98 there, because the corpus was purpose-written and single-concept — the same lever Hadetskyi et al. [8] claim. Our contribution is the gate and its position, not a negative result about contrast-set design.

Llama is a measurement arm only. No Llama artifact is distributed, under the licence constraint for modified Llama weights.

Prepared but unrun. A CBRN gate on 670 WMDP positives is fully prepared and was not executed — splits audited disjoint, dry run passed, positive controls reproduce the shipped constants, and the GPU stage never started. We report no number for it.

What this paper leaves out. Two further results are omitted here for space and kept in the long version: an evaluation-protocol ladder showing that the same classifier scores 96–98% target coverage when test rows are drawn from the corpus it was fitted on and 38–66% when they come from an independently assembled corpus, and the hate category discussed in §7. The first is a contribution about evaluation methodology rather than about the gate; every coverage figure reported here is already on the separated side of it.

9 Reproducibility and data handling

The verified proof-of-concept implementation, runtime gate server and shipped operating-point kits are available at `https://github.com/kncshw/resgate`. Every number traces to a record directory under `ablation/records/`, each containing the as-run scripts, the logs, the per-row outputs and a summary JSON with sha256 for every input; the record names are given beside the results they back, and the long version of this paper carries the complete index. Operating points ship as compact `.pt` files — a direction blob and an operating point carrying its own provenance. Positive controls are mandatory and gated: no result is reported unless the implementation first reproduces six published constants to float64.

Data handling. The WMDP prompts used for the bio work were shared privately by the RepIt authors, whose paper records them as not publicly available pending gated access. They are not redistributed, no weight, direction blob or operating-point file from the bio arm ships, and **no prompt text from that collection appears anywhere in this paper**. Prompt sets are described by name, source and count only. Every

artifact bound for the bio record was screened against 56,004 distinct four-word needles drawn from all 2954 author prompt strings; escalating to 58,362 eight-word needles over all 121 files in `records/bio-m1-live-serve-2026-10-15/` gives 0 hits.

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